

Code-Layout, Readability and Reusability

Date	07 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Python indentation (PEP 8) followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, dataset, model, and application code.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as <code>loan_amount</code> , <code>applicant_income</code> , <code>credit_history</code> , and <code>prediction_result</code> improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>predict_loan()</code> , <code>preprocess_data()</code> , and <code>load_model()</code> clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments for preprocessing, model loading, prediction, and Flask routes.

6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, model prediction, and Flask application improve maintainability.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated logic is minimized using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling ensure reliable application execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	Dataset cleaning, feature engineering	High
2	Machine Learning Model	Scikit-learn	Loan eligibility prediction	High
3	Flask Application	Flask (Python)	Web application routing and prediction	High
4	HTML Templates	HTML5, Bootstrap 5	User interface pages	Medium
5	Prediction Utility Functions	Python	Model loading and prediction workflow	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5/5	Well-organized project with a clear folder hierarchy.
Readability	5/5	Meaningful naming conventions, consistent formatting, and comments improve readability.
Reusability	5/5	Modular functions and reusable components simplify maintenance and future enhancements.
Documentation / Comments	5/5	Code is well documented, making it easy to understand and extend.
Overall Score	5/5	The project follows software engineering best practices and maintains high standards of code quality, readability, and reusability.